MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU -UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 Admission onwards)
Minor Course – MG3DSCBCH202 – Techniques in Biochemistry and Forensic Science

Time: 1.5 Hours Max. Marks: 50

Part A
Multiple Choice Questions
(Answer all questions; each question carries 1mark)

1. Biochemical techniques are important in [U][1]
   A). Understanding biological processes B). Disease diagnosis C) Therapeutic development D) All of the above

2. Which among the following is a column chromatography procedure [K][1]
   A). Affinity chromatography B). Gel Exclusion Chromatography C) Both (a) and (b) D) Paper chromatography

3. Colorimeter measures the intensity of light in the region of the spectrum [K][2]
   A) 400 – 750 nm B) 160 – 350 nm C) 160 – 700nm D) All of the above

4. Polymerase Chain Reaction involves the steps [U][3]
   A) Denaturation B) Annealing C) Extension D) All of the above

5. Ultra-sensitive detection is facilitated using [U][3]
   a) Chromatography b) RIA c) ELISA d) Mass Spectrometry

(5 x 1 = 5 marks)

Part B
Fill in the Blanks
(Answer all questions; each question carries 2 marks)

6. The term chromatography was coined by…… [K][1]

7. ……. law governs absorption of light by a given solute in solution [K][2]

8. Centrifugation works on the principle of …… [U][2]

9. …… is a most reliable and commonly used forensic DNA source [U][3]

10. DNA finger printing analyzes specific regions in the DNA called …… [U][3]

(5 x 2 = 10 marks)

Part C
Short Answer Questions
(Answer any 5 questions; each question carries 3 marks)

11. Explain the significance of distribution coefficient. [U][1]

12. Discuss the different types of PAGE [K][1]

13. Explain Beer Lambert's law. [K][2]

14. Differentiate between simple and compound microscope. [U][2]

15. Explain the function and types of electron microscope. [U][2]

16. Discuss the various applications of ELISA. [U][3]

17. Illustrate the significance of maternity testing. [An][3]

(5 x 3 = 15 marks)

Part D

Short Essay Type Questions

(Answer any 4 questions; each question carries 5 marks)

18. Evaluate the technique of Thin Layer Chromatography, highlighting its principle, methodology, advantages, and applications. [E][1]

19. Explain Southern blotting. [U][1]

20. Explain the procedure of spectrophotometry. [U][2]

21. Explain how you would use the technique of centrifugation to purify biological molecule [A][2]

22. Analyze the process of PCR based on the steps involved [An][3]

23. Describe the procedure of DNA finger printing. [K][3]

(4x 5 = 20 marks)